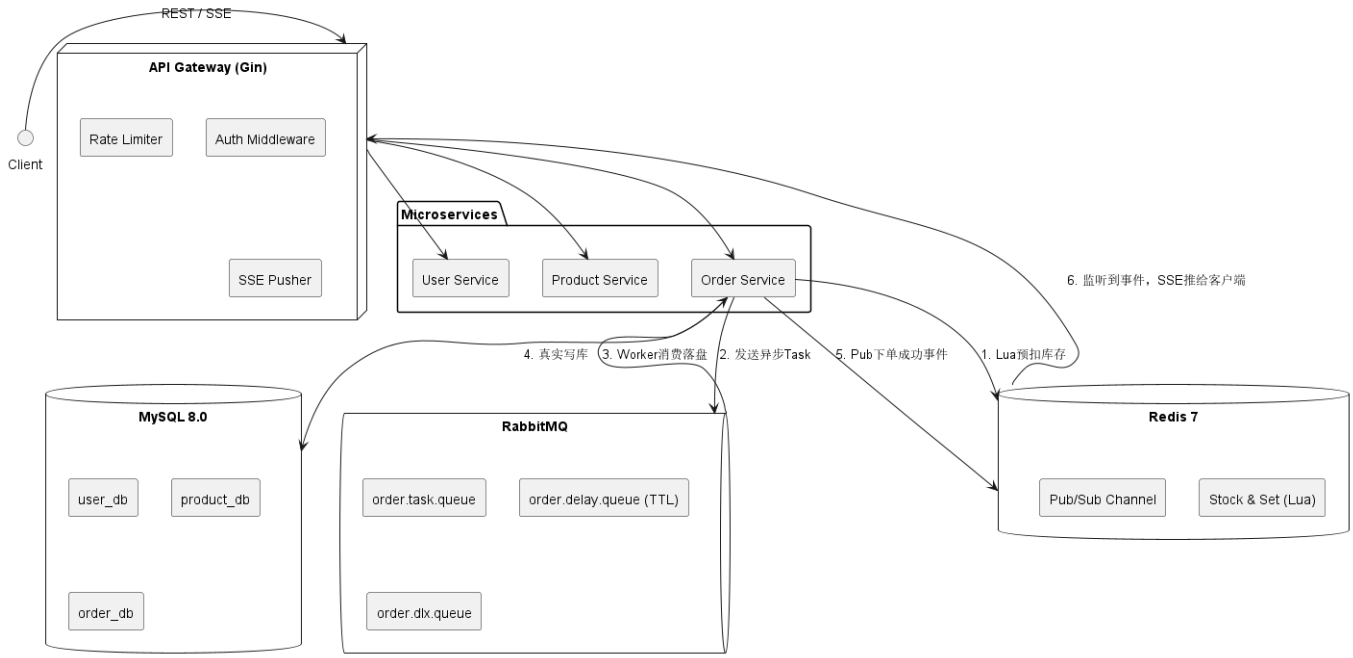
# 在线盲盒与潮玩秒杀系统 - SDD

## 1. 开发环境与技术栈

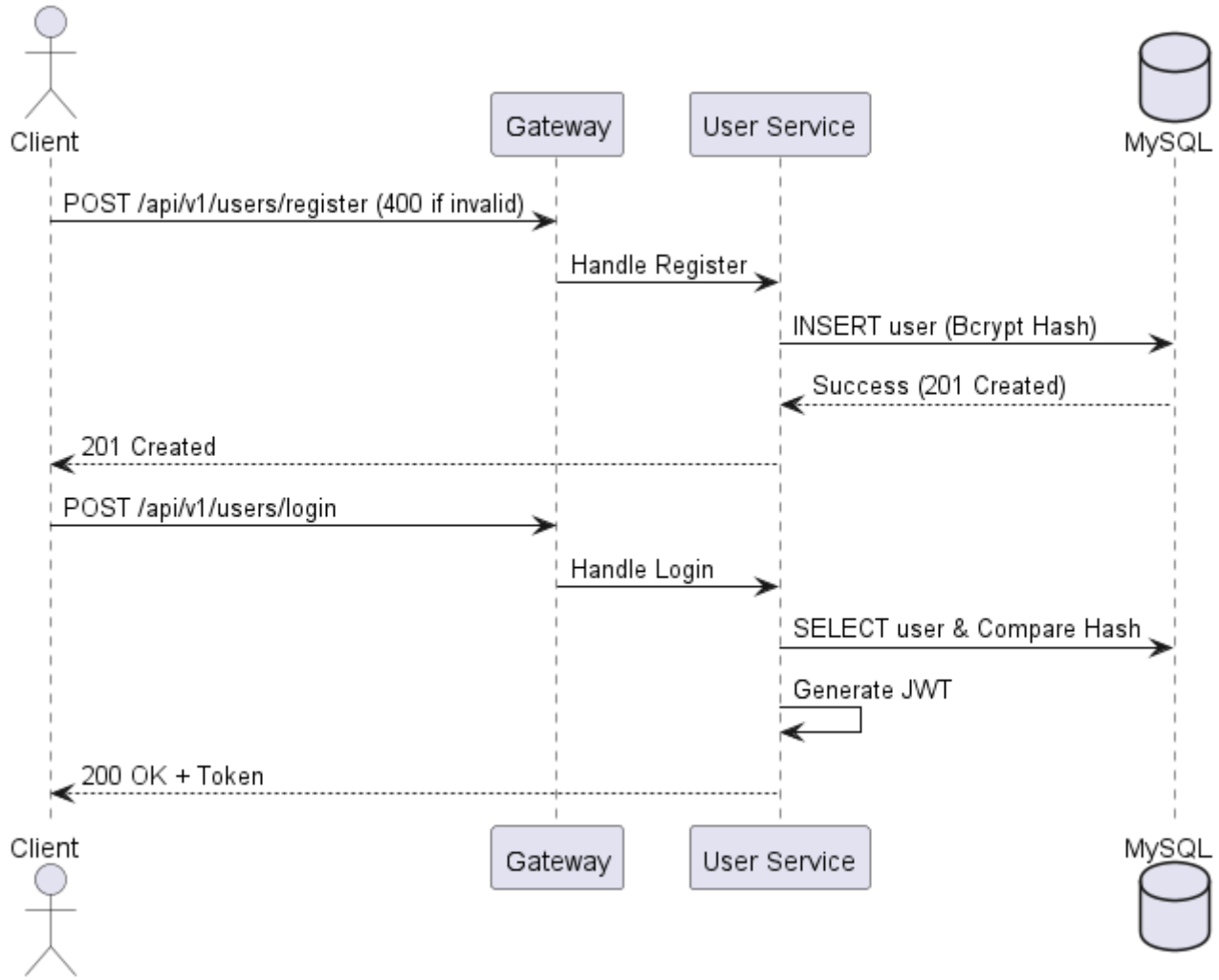
- **操作系统**: WSL2 (Ubuntu)
- **编程语言**: Go 1.21+ (利用 Goroutine + Channel 实现高并发协同)
- **Web 框架**: Gin (高性能路由与中间件)
- **ORM**: GORM (对接 MySQL 8.0)
- **缓存中间件**: Redis 7 (Lua 脚本防超卖，Pub/Sub 用于长连接事件通知)
- **消息队列**: RabbitMQ (基于 Docker 部署，使用 Topic Exchange 与 DLX)
- **架构风格**: 基于 Monorepo 的轻量级微服务架构

## 2. 架构图 (Architecture Diagram)

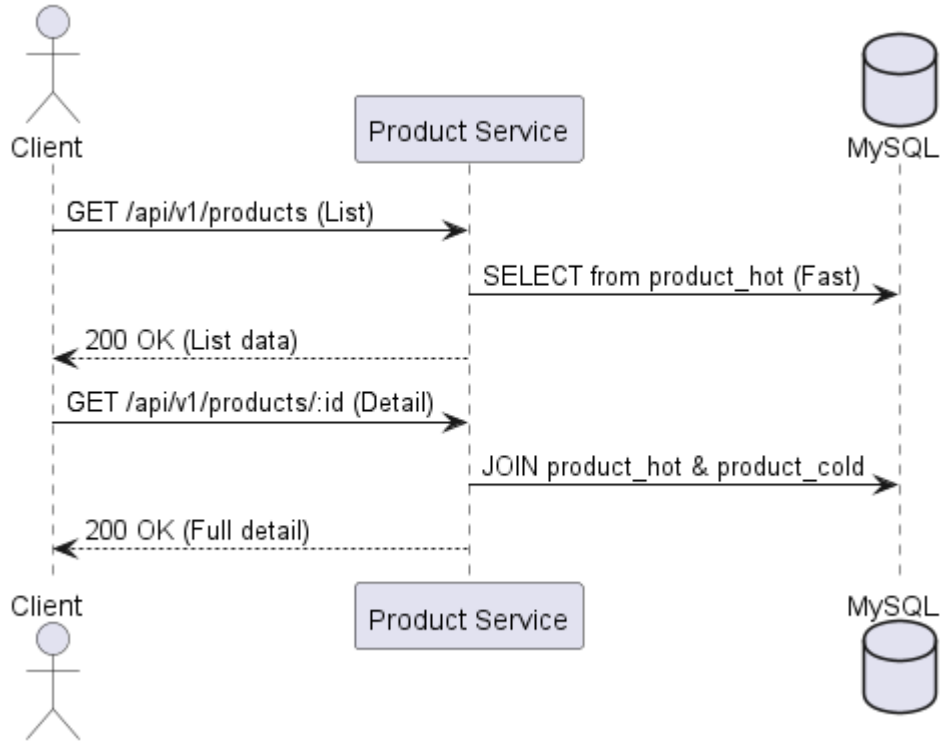



## 3. 微服务交互时序图 (Sequence Diagrams)

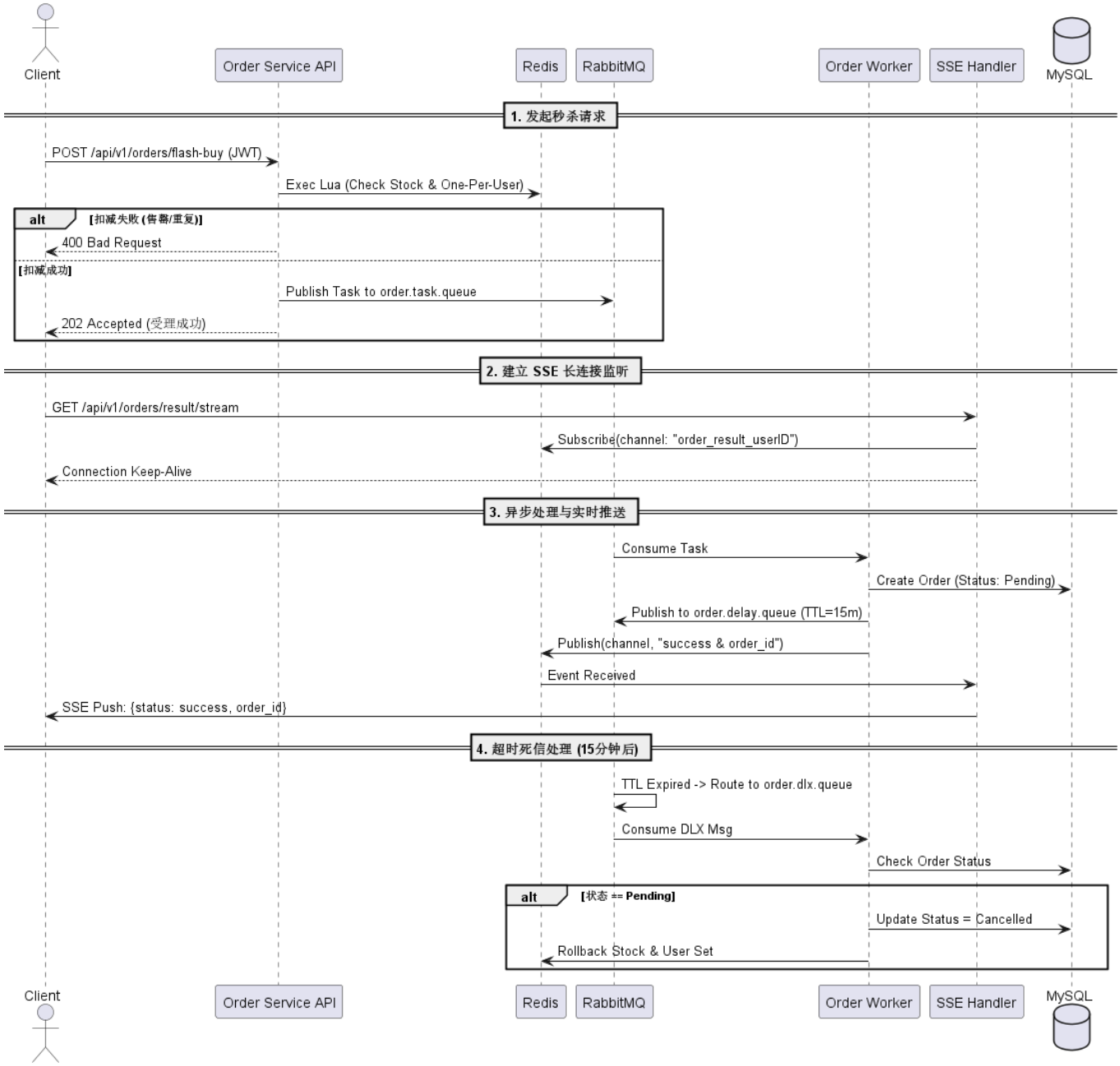
### 3.1 User Service (注册与登录)




3.2 Product Service (高频查询与垂直分表)




3.3 Order Service (核心链路：预扣减 + MQ + SSE 推送 + DLX)


Client
Order Service API
Redis
RabbitMQ
Order Worker
SSE Handler
MySQL
1. 发起秒杀请求
POST /api/v1/orders/flash-buy (JWT)
Exec Lua (Check Stock & One-Per-User)
alt
[扣减失败(售罄/重复)]
400 Bad Request
[扣减成功]
Publish Task to order.task.queue
202 Accepted (受理成功)
2. 建立 SSE 长连接监听
GET /api/v1/orders/result/stream
Subscribe(channel: "order_result_userID")
Connection Keep-Alive
3. 异步处理与实时推送
Consume Task
Create Order (Status: Pending)
Publish to order.delay.queue (TTL=15m)
Publish(channel, "success & order_id")
Event Received
SSE Push: {status: success, order_id}
4. 超时死信处理 (15分钟后)
TTL Expired -> Route to order.dlx.queue
Consume DLX Msg
Check Order Status
alt
[状态 == Pending]
Update Status = Cancelled
Rollback Stock & User Set
Client
Order Service API
Redis
RabbitMQ
Order Worker
SSE Handler
MySQL